Chapter 5

Oscillations

I covered this chapter in 4.5 *fifty-minute lectures.*

The first two sections of this chapter deal with the one-dimensional simple harmonic oscillator, something almost all our students are familiar with. On the other hand, I found many students distressingly weak at the trigonometry behind the various ways to write oscillations — hence the rather ponderous enumeration of the four equivalent forms for an oscillatory solutions in Section 5.2. Section 5.3 is about the two-dimensional oscillator, which is obviously important and comes up from time to time later in the book but plays no role in the rest of the chapter; so you could omit this section.

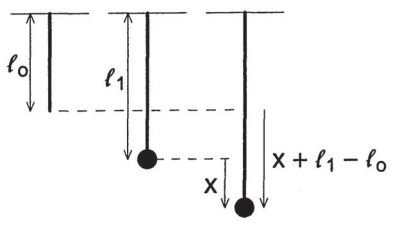
Section 5.4 is about the damped oscillator (back in one dimension) and 5.5 and 5.6 treat the driven damped oscillator and resonance — topics that every physics major should surely understand and that are essential if you're going to cover Chapter 12 on the chaos of non-linear oscillators. Finally, Sections 5.7–5.9 use Fourier series to solve for the motion of an oscillator that is driven by an arbitrary periodic force. While these three sections could certainly be omitted, my class voted overwhelmingly to include them.

There are several demonstration experiments you can show on driven damped oscillations, including the very simple on described in footnote 14.

Solutions to Problems for Chapter 5

5.1 ⋆ **(a)** When the spring is at the new equilibrium position it is extended by an amount $l_1 - l_o$, so its tension is $k(l_1 - l_o)$. This must balance the weight mg. Therefore, $k(l_1 - l_o) = mg$. When it is stretched a further x, its total extension is $x + l_1 - l_o$ and the tension is $k(x + l_1 - l_o)$ upward. Thus the total downward force on the mass is

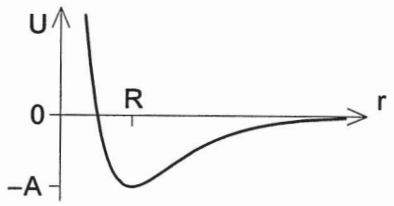
$$F = -k(x + l_1 - l_o) + mg = -kx.$$



(b) The total PE is $U = U_{\text{sp}} + U_{\text{gr}}$ or

$$U = \tfrac{1}{2}k(x + l_1 - l_o)^2 - mgx = \tfrac{1}{2}kx^2 + k(l_1 - l_o)x - mgx + \tfrac{1}{2}k(l_1 - l_o)^2 = \tfrac{1}{2}kx^2 + \text{const}$$

5.2 ⋆ When $r = 0$, $U = A[(e^{R/S} - 1)^2 - 1]$, which is large and positive since $R \gg S$. When $r \to \infty$, U is negative and approaches 0. The smallest possible value of U is when $r = R$ and $U = -A$; that is, the equilibrium separation is $r_o = R$. If we set $r = R + x$ and make a Taylor expansion of the exponential term in U, then

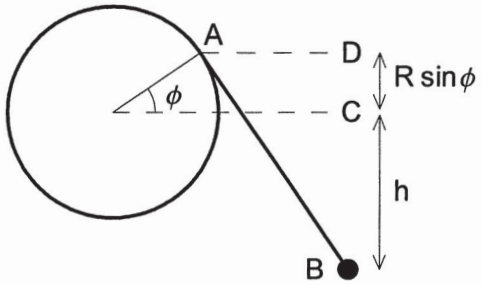


$$U = A\left[\left(\left\{1 - \frac{x}{S} + \cdots\right\} - 1\right)^2 - 1\right] \approx -A + A\left(\frac{x}{S}\right)^2 = \text{const} + \tfrac{1}{2}kx^2$$

where $k = 2A/S^2$.

5.3 ⋆ The height of the mass below the pivot is $l\cos\phi$. Therefore the height above the bottom is $l(1 - \cos\phi)$ and the PE is $U = mgl(1 - \cos\phi)$. If ϕ is small, $\cos\phi \approx 1 - \frac{1}{2}\phi^2$ and $U \approx \frac{1}{2}mgl\phi^2 = \frac{1}{2}k\phi^2$, where $k = mgl$.

5.4 ⋆⋆ The PE is $U = -mgh$ where h is the height of the mass, measured down from the level of the cylinder's center. To find h, note first that as the pendulum swings from equilibrium to angle ϕ, a length $R\phi$ of string unwinds from the cylinder. Thus the length of string away from the cylinder is $AB = (l_o + R\phi)$, and the height BD is $BD = (l_o + R\phi)\cos\phi$. Since the height $CD = R\sin\phi$, we find by subtraction that $h = BD - CD = l_o\cos\phi + R(\phi\cos\phi - \sin\phi)$. Therefore



$$U = -mgh = -mg[l_o\cos\phi + R(\phi\cos\phi - \sin\phi)].$$

If ϕ remains small we can write $\cos\phi \approx 1 - \phi^2/2$ and $\sin\phi \approx \phi$, to give

$$U \approx -mg\left\{l_o - \tfrac{1}{2}l_o\phi^2 + R\left[\phi(1 - \tfrac{1}{2}\phi^2) - \phi\right]\right\} \approx -mgl_o + \tfrac{1}{2}mgl_o\phi^2 = \text{const} + \tfrac{1}{2}k\phi^2$$

where in the third expression I dropped the term in ϕ^3. The constant $k = mgl_o$, which is the same as for a simple pendulum of length l_o. Evidently, wrapping the string around a cylinder makes no difference for small oscillations.

5.5 ⋆ **(a)** In the form (I), $x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$, we can replace the exponentials using Euler's formula $e^{\pm i\omega t} = \cos\omega t \pm i \sin\omega t$ to give

$$x(t) = B_1 \cos\omega t + B_2 \sin\omega t \qquad \text{(II)}$$

where $B_1 = C_1 + C_2$ and $B_2 = i(C_1 - C_2)$.

(b) To get form (III) from form (II), define A and δ to be the hypotenuse and lower angle of a right triangle with base B_1 and height B_2 as in Figure 5.4. Then (II) can be rewritten as in Equation (5.11) to give the form (III).

(c) Since $\cos\theta = \mathrm{Re}\, e^{i\theta}$, we can rewrite the form (III) as

$$x(t) = \mathrm{Re}\, A e^{i(\omega t - \delta)} = \mathrm{Re}\, C e^{i\omega t} \qquad \text{(IV)}$$

where $C = Ae^{-i\delta}$.

(d) Finally, since $\mathrm{Re}\, z = \frac{1}{2}(z + z^*)$, we can rewrite (IV) as

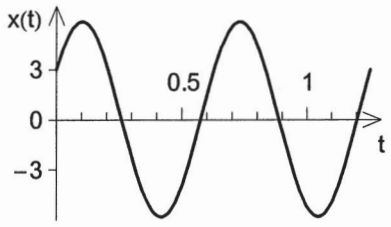
$$x(t) = \tfrac{1}{2}(Ce^{i\omega t} + C^* e^{-i\omega t}) = C_1 e^{i\omega t} + C_2 e^{-i\omega t} \qquad \text{(I)}$$

where $C_1 = C/2$ and $C_2 = C^*/2$.

5.6 ⋆ The position can be written as $x(t) = A\cos(\omega t - \delta)$, where $A = 2x_o$ and we know that $x(0) = x_o$ and $v(0) < 0$. Putting $t = 0$, we see that $\cos(-\delta) = 0.5$ and hence that $\delta = \pm\pi/3$. The velocity is $v(t) = -A\omega\sin(\omega t - \delta)$, so the condition that $v(0) < 0$ implies that $\sin(\delta) < 0$. Therefore $\delta = -\pi/3$ and $x(t) = 2x_o \cos(\omega t + \pi/3)$.

5.7 ⋆ **(a)** Since $x(t) = B_1 \cos\omega t + B_2 \sin\omega t$, $x(0) = B_1$ and $v(0) = \omega B_2$. Therefore, $B_1 = x_o$ and $B_2 = v_o/\omega$.

(b) If $k = 50$ N/m and $m = 0.5$ kg, then $\omega = \sqrt{k/m} = 10\ \mathrm{s}^{-1}$. If $x_o = 3.0$ m and $v_o = 50$ m/s, then $B_1 = 3$ m, $B_2 = 5$ m.



(c) If we rewrite $x(t)$ in the form $x(t) = A\cos(\omega t - \delta)$, with $\delta = \arctan(B_2/B_1) = 1.03$ rad, then x first vanishes when $(\omega t - \delta) = \pi/2$ or $t = 0.26$ s, and v first vanishes when $(\omega t - \delta) = 0$ or $t = \delta/\omega = 0.10$ s.

5.8 ⋆ **(a)** $\omega = \sqrt{k/m} = \sqrt{80/0.2} = 20\ \mathrm{s}^{-1}$, $f = \omega/2\pi = 3.2$ Hz, and $\tau = 2\pi/\omega = 0.31$ s.

(b) Since $x_o = 0$, $A\cos(-\delta) = 0$, so $\delta = \pm\pi/2$. Since $v_o = \omega A \sin\delta = 40$ m/s, δ must be positive, $\delta = +\pi/2$, and therefore $A = v_o/\omega = 2$ m.

5.9 ⋆ We are given that $A = 0.2$ m and $v_o = 1.2$ m/s. We know that $E = \frac{1}{2}kA^2 = \frac{1}{2}mv_o^2$. Therefore $m/k = A^2/v_o^2$ and $\tau = 2\pi\sqrt{m/k} = 2\pi A/v_o = 1.05$ s.

5.10 ⋆ If $F = -F_o \sinh \alpha x$, then $U = -\int F\,dx = (F_o/\alpha)\cosh \alpha x$. The only equilibrium position is at $x = 0$ and, for points close to this, Taylor's series gives

$$U(x) \approx (F_o/\alpha)(1 + \tfrac{1}{2}\alpha^2 x^2) = \tfrac{1}{2}kx^2 + \text{const},$$

where $k = \alpha F_o$. The angular frequency of oscillations is $\omega = \sqrt{k/m} = \sqrt{\alpha F_o/m}$.

5.11 ⋆ The given information gives two expressions of the total energy E

$$E = \tfrac{1}{2}mv_1^2 + \tfrac{1}{2}kx_1^2 \quad \text{and} \quad E = \tfrac{1}{2}mv_2^2 + \tfrac{1}{2}kx_2^2. \tag{i}$$

Equating these two, we find $m(v_1^2 - v_2^2) = k(x_2^2 - x_1^2)$. This implies that

$$\omega^2 = \frac{k}{m} = \frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}.$$

We know also that $E = \frac{1}{2}kA^2$, and inserting this in the first of Eqs.(i) we conclude that

$$A^2 = \frac{m}{k}v_1^2 + x_1^2 = \frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}v_1^2 + x_1^2 = \frac{x_2^2 v_1^2 - x_1^2 v_2^2}{v_1^2 - v_2^2}.$$

5.12 ⋆⋆ Because $\sin^2(\omega t - \delta)$ oscillates symmetrically between 0 and 1, its average over a cycle is fairly obviously $\frac{1}{2}$. To prove it, write $\sin^2\theta = \frac{1}{2}[1 - \cos 2\theta]$, so that

$$\langle \sin^2(\omega t - \delta)\rangle = \frac{1}{2\tau}\int_0^\tau [1 - \cos 2(\omega t - \delta)]dt = \frac{1}{2} - \frac{1}{2\tau}\big[\sin 2(\omega t - \delta)\big]_0^\tau = \frac{1}{2}$$

where the final square bracket is zero because the sine function is τ-periodic. The corresponding result with a cosine follows in exactly the same way.

We know from (5.16) that $E = \frac{1}{2}kA^2$, and, since $T = \frac{1}{2}kA^2\sin^2(\omega t - \delta)$, it follows that $\langle T\rangle = \frac{1}{4}kA^2 = \frac{1}{2}E$, and similarly for $\langle U\rangle$.

5.13 ⋆⋆ Since $U(r) = U_o(r/R + \lambda^2 R/r)$, its derivative is $dU/dr = U_o(1/R - \lambda^2 R/r^2)$, which vanishes at $r = \lambda R$ and nowhere else. Clearly $U(r) \to +\infty$ when $r \to 0$ or ∞, so $U(r)$ has a minimum at $r = r_o = \lambda R$. If we write $r = r_o + x$ then

$$\begin{aligned} U = \lambda U_o\left(\frac{r_o + x}{r_o} + \frac{r_o}{r_o + x}\right) &= \lambda U_o\left(1 + \frac{x}{r_o} + \left[1 + \frac{x}{r_o}\right]^{-1}\right) \\ &\approx \lambda U_o\left(1 + \frac{x}{r_o} + 1 - \frac{x}{r_o} + \frac{x^2}{r_o^2}\right) = \lambda U_o\left(2 + \frac{x^2}{\lambda^2 R^2}\right). \end{aligned}$$

where, in the second line, I dropped all terms in $(x/r_o)^3$ and higher. This has the expected form $U = \frac{1}{2}kx^2 + \text{const}$, where $k = 2U_o/(\lambda R^2)$. The angular frequency is $\omega = \sqrt{k/m} = \sqrt{2U_o/(m\lambda R^2)}$.

5.14 ⋆ Given that $\mathbf{F} = -(k_x x, k_y y)$, the PE is

$$U(\mathbf{r}) = -\int_0^{\mathbf{r}} \mathbf{F}(\mathbf{r}')\cdot d\mathbf{r}' = \int_0^{\mathbf{r}} (k_x x'dx' + k_y y'dy') = k_x \int_0^x x'dx' + k_y \int_0^y y'dy' = \tfrac{1}{2}(k_x x^2 + k_y y^2).$$

5.15 ⋆ If we replace the variable t by $t' = t + t_o$, Eq. (5.19) becomes

$$x = A_x \cos(\omega t' - \omega t_o - \delta_x) \quad \text{and} \quad y = A_y \cos(\omega t' - \omega t_o - \delta_y)$$

and if we then choose t_o such that $\omega t_o = -\delta_x$ these become

$$x = A_x \cos(\omega t') \quad \text{and} \quad y = A_y \cos(\omega t' - [\delta_y - \delta_x]).$$

If we rename t' as t and set $\delta_y - \delta_x = \delta$, this is the desired form.

5.16 ⋆ With $\delta = \pi/2$ Eq.(5.20) reads

$$x = A_x \cos(\omega t) \quad \text{and} \quad y = A_y \cos(\omega t - \pi/2) = A_y \sin(\omega t)$$

from which it follows that $x^2/A_x^2 + y^2/A_y^2 = \cos^2(\omega t) + \sin^2(\omega t) = 1$, the equation of an ellipse with semi-major and semi-minor axes A_x and A_y.

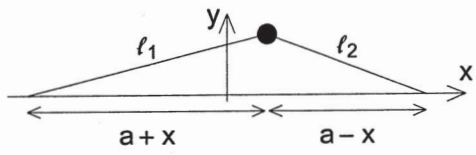
5.17 ⋆⋆ **(a)** Suppose that the ratio of frequencies is rational, that is $\omega_x/\omega_y = p/q$, where p and q are integers. Then let $\tau = 2\pi p/\omega_x = 2\pi q/\omega_y$. Now consider the following

$$x(t+\tau) = A_x \cos[\omega_x(t+\tau]) = A_x \cos[\omega_x t + 2\pi p] = A_x \cos[\omega_x t] = x(t)$$

where in the second equality I used our definition of τ and in the second the fact that if p is an integer then $\cos(\theta + 2\pi p) = \cos(\theta)$. This shows that $x(t)$ is periodic with period τ. By exactly the same argument, $y(t)$ is also periodic with the same period τ, and we've proved that the whole motion is likewise. What we usually call *the* period of the motion is the value of $\tau = 2\pi p/\omega_x$ with p and q the *smallest* integers for which $\omega_x/\omega_y = p/q$.

(b) Suppose the motion is periodic. Then there is a τ such that $x(t+\tau) = x(t)$ and $y(t+\tau) = y(t)$. Running the previous argument backward, we see that $\omega_x\tau$ must be an integer multiple of 2π, that is $\omega_x\tau = 2\pi p$ for some integer p. Similarly $\omega_y\tau = 2\pi q$ for some integer q. Dividing these two conclusions, we see that $\omega_x/\omega_y = p/q$ and the ratio of frequencies is rational. Therefore, if the ratio is irrational, the motion cannot be periodic.

5.18 ⋆⋆⋆ When the mass is at position (x, y), the lengths of the two springs are l_1 and l_2, where



$$l_1 = \sqrt{(a+x)^2 + y^2} = a\left(1 + \frac{2x}{a} + \frac{x^2+y^2}{a^2}\right)^{1/2}$$

$$\approx a\left[1 + \frac{1}{2}\left(\frac{2x}{a} + \frac{x^2+y^2}{a^2}\right) - \frac{1}{8}\left(\frac{2x}{a}\right)^2\right] = a + x + \frac{y^2}{2a}.$$

Here, in passing from the first to the second line, I have used the Taylor expansion $(1+\epsilon)^{1/2} = 1 + \frac{1}{2}\epsilon - \frac{1}{8}\epsilon^2 + \cdots$, dropping all terms of third degree in x or y, but being careful to keep all terms of second degree. The PE of spring 1 is therefore

$$\begin{aligned} U_1 = \tfrac{1}{2}k(l_1 - l_o)^2 &= \tfrac{1}{2}k[(a - l_o) + x + y^2/2a]^2 \\ &\approx \tfrac{1}{2}k[(a - l_o)^2 + 2(a - l_o)x + x^2 + (1 - l_o/a)y^2] \end{aligned}$$

where, again, I have dropped terms of degree three in x or y. To find U_2, we have only to replace x by $-x$, and for the total PE we just add U_1 and U_2. When we do this, the terms linear in x cancel, leaving

$$U = U_1 + U_2 = k[x^2 + (1 - l_o/a)y^2] + \text{const}$$

which has the form (5.104), apart from the unimportant constant.

If $a < l_o$, the coefficient of y^2 is negative and the equilibrium at the origin O is unstable. This is because, with $a < l_o$, the springs are in compression at O. When the mass moves a little from O along the y axis, the compression in the springs forces it further away, causing the instability.

5.19 ⋆⋆⋆ The simplest way to find the total PE of all four springs is treat them two at a time. The two springs anchored on the x axis constitute the system of Problem 5.18, for which we found that

$$U_1 + U_2 = k[x^2 + (1 - l_o/a)y^2].$$

(See the solution to that problem. I've dropped an uninteresting constant here.) In exactly the same way, the PE of the two springs anchored on the y axis is

$$U_3 + U_4 = k[y^2 + (1 - l_o/a)x^2].$$

Adding these, we find for the total PE of all four springs

$$U = U_1 + U_2 + U_3 + U_4 = k[x^2 + y^2 + (1 - l_o/a)(x^2 + y^2)] = k\frac{2a - l_o}{a}r^2,$$

where I've used the fact that $x^2 + y^2 = r^2$. This has the advertised form $U = \frac{1}{2}k'r^2$ with an effective spring constant $k' = 2k(2a - l_o)/a$. The corresponding force is $\mathbf{F} = -\boldsymbol{\nabla}U = -k'\mathbf{r}$.

5.20 ⋆ The derivative of the decay parameter, $\beta - \sqrt{\beta^2 - \omega_o^2}$, is $1 - \beta/\sqrt{\beta^2 - \omega_o^2}$. With $\beta > \omega_o$ the second term is clearly greater than 1, so the derivative is negative, and the decay parameter *decreases* as β increases. The graph is shown in Fig.5.13.

5.21 ⋆ If $x = te^{-\beta t}$, then

$$\dot{x} = (1 - \beta t)e^{-\beta t} \quad \text{and} \quad \ddot{x} = (\beta^2 t - 2\beta)e^{-\beta t}.$$

Therefore

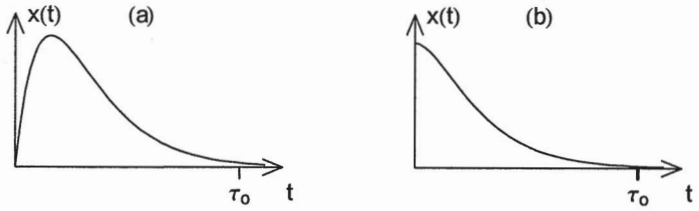
$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = \left[(\beta^2 t - 2\beta) + 2\beta(1 - \beta t) + \omega_o^2 t\right] e^{-\beta t} = [\omega_o^2 - \beta^2]e^{-\beta t} = 0$$

if (and only if) $\beta = \omega_o$. That is, $x = te^{-\beta t}$ *does* satisfy Eq.(5.28) if the damping is critical.

5.22 ⋆ **(a)** The general solution for a critically damped oscillator ($\beta = \omega_o$) is given in (5.44) as $x(t) = e^{-\omega_o t}(C_1 + C_2 t)$. Thus

$$x_o = x(0) = C_1 \qquad \text{and} \qquad v_o = \dot{x}(0) = C_2 - \omega_o C_1 \,. \tag{ii}$$

Here $x_o = 0$, so $C_1 = 0$ and $C_2 = v_o$. Therefore, $x(t) = v_o t e^{-\omega_o t}$.



(b) In this case $v_o = 0$ and Eqs.(ii) imply that $C_1 = x_o$ and $C_2 = \omega_o x_o$. Therefore $x(t) = x_o e^{-\omega_o t}(1 + \omega_o t)$. When $t = \tau_o$, the natural period, $x = x_o e^{-2\pi}(1 + 2\pi) = 0.0136 x_o$. The motion is almost 99% damped out.

5.23 ⋆ Because $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$,

$$\frac{dE}{dt} = m\dot{x}\ddot{x} + kx\dot{x} = \dot{x}(m\ddot{x} + kx) = \dot{x}(-b\dot{x}) = vF_{\text{dmp}}$$

where, for the third equality, I used the differential equation (5.24). Since vF_{dmp} is the rate at which F_{dmp} does work on the oscillator, this is the requested result.

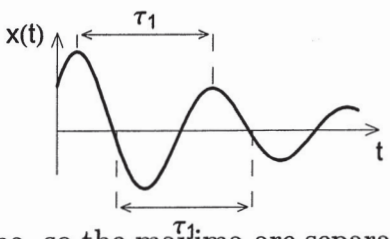
5.24 ⋆ As long as $\beta < \omega_o$, we can define $\omega_1 = \sqrt{\omega_o^2 - \beta^2}$ and we can use as two independent solutions

$$x_1(t) = e^{-\beta t}\cos(\omega_1 t) \qquad \text{and} \qquad x_2(t) = e^{-\beta t}\sin(\omega_1 t).$$

If $\beta \to \omega_o$, then $\omega_1 \to 0$ and so $x_1(t) \to e^{-\beta t}$. That is, as $\beta \to \omega_o$, our first solution for the case $\beta < \omega_o$ becomes the first solution for the case $\beta = \omega_o$.

Unfortunately, in the same limit, $x_2(t) \to 0$, and our second solution evaporates. However, for our second solution we could equally have used $\tilde{x}_2(t) = e^{-\beta t}\sin(\omega_1 t)/\omega_1$. Since $\sin(kt)/k \to t$ as $k \to 0$, this new second solution does not vanish as $\omega_1 \to 0$; instead, as $\beta \to \omega_o$, the new solution satisfies $\tilde{x}_2(t) \to te^{-\beta t}$ and we obtain the second solution for the case $\beta = \omega_o$.

5.25 ⋆⋆ **(a)** Because $x(t) = Ae^{-\beta t}\cos(\omega_1 t - \delta)$, its derivative is $dx/dt = -Ae^{-\beta t}[\beta\cos(\cdots) + \omega_1\sin(\cdots)]$. The maxima and minima of $x(t)$ occur when this derivative vanishes, that is, when $\tan(\omega_1 t - \delta) = -\beta/\omega_1$. Because $\tan\theta$ is π-periodic, the zeroes of dx/dt are equally spaced, with separation π/ω_1. The zeroes of the derivatives correspond alternately to maxima and minima, so the maxima are separated by a time $\tau_1 = 2\pi/\omega_1$.



(b) The zeroes of $x(t)$ occur when $\cos(\omega_1 t - \delta) = 0$. Thus they are regularly spaced with separation of π/ω_1, which equals $\tau_1/2$.

(c) With $\beta = \omega_o/2$, the amplitude shrinks by a factor

$$e^{-\beta\tau_1} = e^{-2\pi\beta/\sqrt{\omega_o^2-\beta^2}} = e^{-2\pi/\sqrt{3}} = 0.027$$

(This is much more shrinkage than in the picture, for which β was chosen to be $\omega_o/10$ and the shrinkage factor is about 0.53).

5.26 ⋆⋆ The damping changes the frequency to $\omega_1 = \sqrt{\omega_o^2 - \beta^2}$, which we can solve to give

$$\beta = \omega_o\sqrt{1 - \frac{\omega_1^2}{\omega_o^2}} = \omega_o\sqrt{1 - \frac{\tau_o^2}{\tau_1^2}} = \omega_o\sqrt{1 - 0.998} = 0.0447\omega_o = 0.281 \text{ s}^{-1}$$

After a time $t = 10\tau_1 \approx 10\tau_o$, the amplitude will have changed by a factor of

$$e^{-\beta t} \approx e^{-10\beta\tau_o} = e^{-20\pi\beta/\omega_o} = e^{-20\pi(0.0447)} = 0.060\,.$$

In other words, the amplitude will have diminished by a factor of $1/0.060 = 17$. Clearly the change of amplitude of by a factor of 17 is far more noticeable than the change of period by 0.1%.

5.27 ⋆⋆ The question is: How many times can the function $x(t)$ vanish? If the oscillator is weakly damped ($\beta < \omega_o$), then according to Eq.(5.38) $x(t)$ contains a factor $\cos(\omega t - \delta)$, which vanishes infinitely many times as it oscillates.

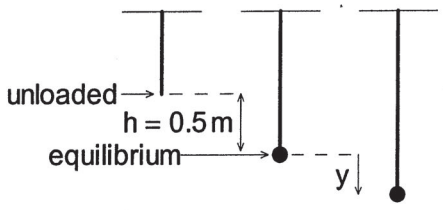
(a) If the oscillator is critically damped ($\beta = \omega_o$), then, according to (5.44), $x(t) = e^{-\beta t}(C_1 + C_2 t)$. This vanishes if and only if $t = -C_1/C_2$; therefore, $x(t)$ vanishes at most once. (It may never vanish — for example, if the motion starts at $t = 0$ and $-C_1/C_2 < 0$.)

(b) If the oscillator is overdamped ($\beta > \omega_o$), then according to (5.40),

$$x(t) = e^{-\beta t}(C_1 e^{\lambda t} + C_2 e^{-\lambda t}) = e^{-(\beta-\lambda)t}(C_1 + C_2 e^{-2\lambda t})$$

where $\lambda = \sqrt{\beta^2 - \omega_o^2}$. This vanishes if and only if $e^{-2\lambda t} = -C_1/C_2$, and, since $e^{-2\lambda t}$ is a monotonic function (always decreasing), this happens at most once.

5.28 ⋆⋆ The final resting position is the equilibrium position, at a height $h = 0.5$ m below the unloaded position. This height is determined by the condition $kh = mg$, from which we see that $\omega_o^2 = k/m = g/h$.



In discussing the oscillations, it is simplest to measure the mass's position $y(t)$ from its equilibrium position. The initial conditions are $y(0) = -h$ and $\dot{y}(0) = 0$. Since the damping is critical, $\beta = \omega_o$ and, according to Equation (5.44),

$$y(t) = (C_1 + C_2 t)e^{-\omega_o t} = -h(1 + \omega_o t)e^{-\omega_o t},$$

because the initial conditions require that $C_1 = -h$ and that $-\omega_o C_1 + C_2 = 0$. Putting in $h = 0.5$ m and $t = 1$ s, we get $\omega_o = \sqrt{g/h} = \sqrt{19.6} = 4.43\ \text{s}^{-1}$ and then

$$y(t) = -(0.5\ \text{m})(1 + 4.43)e^{-4.43} = -0.032\ \text{m} = -3.2\ \text{cm}.$$

That is, it is just 3.2 cm above its final resting position.

5.29 ⋆⋆ After one period, the amplitude has shrunk by a factor $e^{-\beta\tau_1} = 0.5$. Therefore $\beta\tau_1 = \ln 2$. Meanwhile, because $\tau_1 = 2\pi/\omega_1 = 2\pi/\sqrt{\omega_o^2 - \beta^2}$, it follows that $\beta\tau_1 = 2\pi/\sqrt{(\omega_o/\beta)^2 - 1}$. Equating these two expressions for $\beta\tau_1$ and solving, we find $\beta/\omega_o = 1/\sqrt{1 + (2\pi/\ln 2)^2} = 0.1097$. That is, $\beta = 0.11\omega_o$.

$$\frac{\tau_1}{\tau_o} = \frac{\omega_o}{\omega_1} = \frac{\omega_o}{\sqrt{\omega_o^2 - \beta^2}} = \frac{1}{\sqrt{1 - \beta^2/\omega_o^2}} \approx 1 + \frac{1}{2}\frac{\beta^2}{\omega_o^2} = 1.006.$$

Therefore $\tau_1 = 1.006$ s.

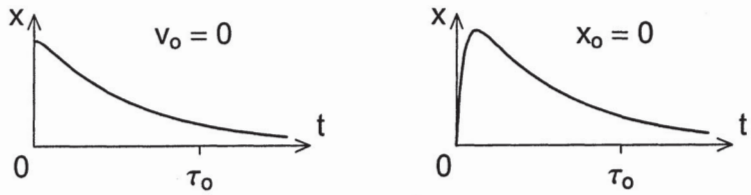
5.30 ⋆⋆ **(a)** From Eq.(5.40) we know that $x = e^{-\beta t}(C_1 e^{\lambda t} + C_2 e^{-\lambda t})$, where $\lambda = \sqrt{\beta^2 - \omega_o^2}$. We can differentiate this to get the velocity v and then set $t = 0$ to give the two equations

$$x_o = C_1 + C_2 \quad \text{and} \quad v_o = \lambda(C_1 - C_2) - \beta(C_1 + C_2)$$

which we can solve to give

$$C_1 = \frac{1}{2\lambda}[x_o(\lambda + \beta) + v_o] \quad \text{and} \quad C_2 = \frac{1}{2\lambda}[x_o(\lambda - \beta) - v_o]$$

(b)



(c) If we let $\beta \to 0$, then $\lambda \to i\omega_o$ and the coefficients C_1 and C_2 become

$$C_1 = \frac{x_o}{2} + \frac{v_o}{2i\omega_o} \quad \text{and} \quad C_2 = \frac{x_o}{2} - \frac{v_o}{2i\omega_o}$$

and our solution becomes

$$x = \frac{x_o}{2}\left(e^{i\omega_o t} + e^{-i\omega_o t}\right) + \frac{v_o}{2i\omega_o}\left(e^{i\omega_o t} - e^{-i\omega_o t}\right) = x_o \cos(\omega_o t) + \frac{v_o}{\omega_o}\sin(\omega_o t)$$

which you should recognize as the general solution for undamped oscillations.

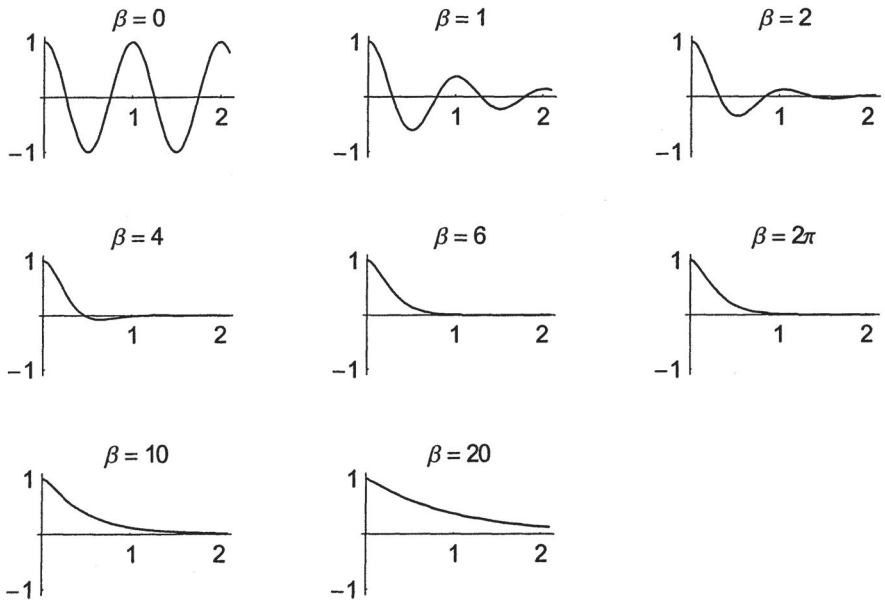
5.31 ⋆⋆ Starting with $\beta < \omega_o$, we know that $x(t)$ has the form (5.37), which we can rewrite as $x(t) = e^{-\beta t}(A\cos\omega_1 t + B\sin\omega_1 t)$, where, as usual, $\omega_1 = \sqrt{\omega_o^2 - \beta^2}$. Given that $x(0) = 1$ and $\dot{x}(0) = 0$, we can solve for A and B and find

$$x(t) = e^{-\beta t}\left(\cos(\omega_1 t) + \frac{\beta}{\omega_1}\sin(\omega_1 t)\right). \qquad \text{(iii)}$$

Given that $\omega_o = 2\pi$, we can use this to plot $x(t)$ for any value of $\beta < \omega_o$. You can also use it for $\beta > \omega_o$. The only complication is that ω_1 is now imaginary, $\omega_1 = i\sqrt{\beta^2 - \omega_o^2} = i\tilde{\omega}_1$, say. If your plotting software is happy to work with complex numbers, you can continue to use exactly the form (iii), but it may be safer to rewrite it in terms of real functions as

$$x(t) = e^{-\beta t}\left(\cosh(\tilde{\omega}_1 t) + \frac{\beta}{\tilde{\omega}_1}\sinh(\tilde{\omega}_1 t)\right). \qquad \text{(iv)}$$

Finally for $\beta = \omega_o$, you almost certainly need to use (5.44) which, for the given initial conditions has the form $x(t) = e^{-\beta t}(1 + \beta t)$. The results are shown below.



5.32 ⋆⋆ **(a)** If we write $x(t)$ in the suggested form, it is easy to see that $x(0) = B_1$ and $\dot{x}(0) = \omega_1 B_2 - \beta B_1$. Therefore, $B_1 = x_o$ and $B_2 = \beta x_o/\omega_1$, so that

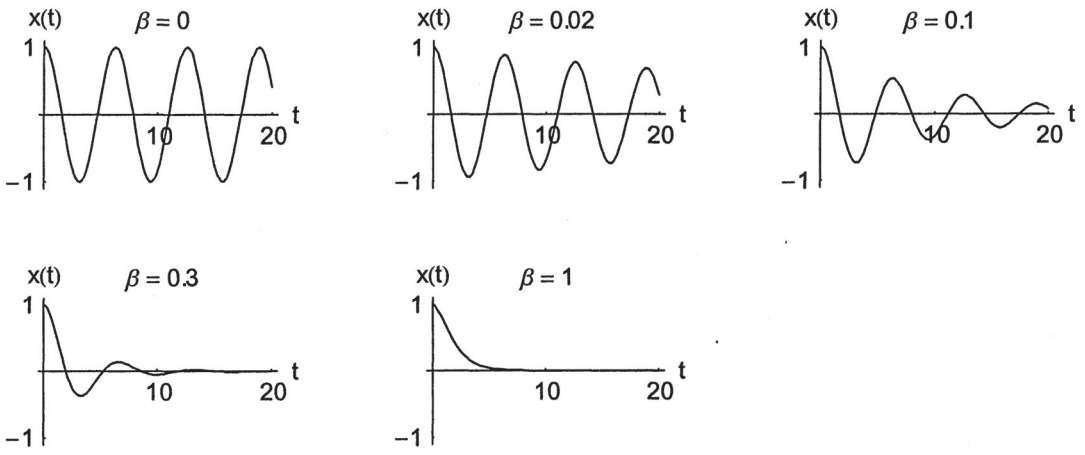
$$x(t) = x_o e^{-\beta t}\left(\cos\omega_1 t + \frac{\beta}{\omega_1}\sin\omega_1 t\right).$$

(b) When $\beta \to \omega_o$, the denominator $\omega_1 = \sqrt{\omega_o^2 - \beta^2}$ approaches zero, but so does the numerator $\sin\omega_1 t$, and $(\sin\omega_1 t)/\omega_1 \to t$. Therefore, the solution of part (a) goes over to

$$x(t) = x_o e^{-\beta t}(1 + \beta t),$$

which is precisely the critical solution (5.44) satisfying the given initial conditions.

(c)



5.33 ⋆ According to Eq.(5.69), $x(t) = A\cos(\omega t - \delta) + e^{-\beta t}[B_1\cos(\omega_1 t) + B_2\sin(\omega_1 t)]$. Setting $t = 0$, we obtain

$$x_o = x(0) = A\cos\delta + B_1 \quad \Longrightarrow \quad B_1 = x_o - A\cos\delta.$$

Similarly, differentiating $x(t)$ and then setting $t = 0$, we find

$$v_o = \dot{x}(0) = A\omega\sin\delta - \beta B_1 + \omega_1 B_2 \quad \Longrightarrow \quad B_2 = \frac{1}{\omega_1}(v_o - A\omega\sin\delta + \beta B_1)$$

as in Eq.(5.70).

5.34 ⋆ We are given that both x_p and x satisfy the same inhomogeneous equation, $Dx_p = f$ and $Dx = f$. Therefore, since D is linear, $D(x - x_p) = Dx - Dx_p = f - f = 0$. That is, the difference $x - x_p = x_h$ is a solution of the homogeneous equation $Dx_h = 0$. Therefore x can always be written as $x = x_p + x_h$ as claimed.

5.35 ⋆⋆ **(a)** $z = x + iy = r\cos\theta + i(r\sin\theta) = r(\cos\theta + i\sin\theta) = re^{i\theta}$.

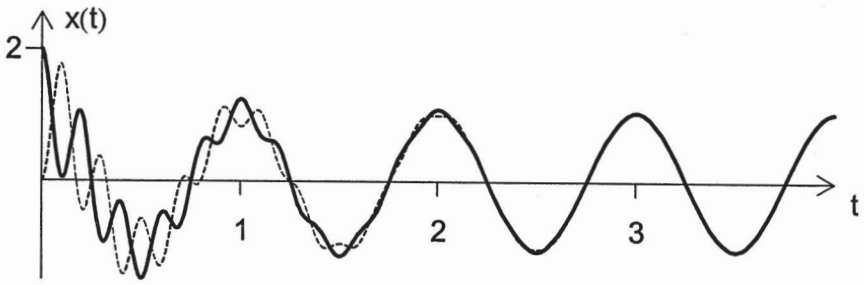
(b) $zz^* = (x + iy)(x - iy) = x^2 + y^2 = |z|^2$.

(c) $z^* = x - iy = r\cos\theta - i(r\sin\theta) = r[\cos(-\theta) + i\sin(-\theta)] = re^{-i\theta}$

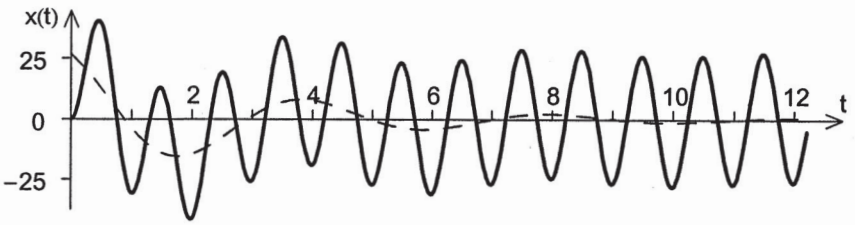
(d) If $z = re^{i\theta}$ and $w = se^{i\phi}$, then $zw = rse^{i(\theta+\phi)}$. Therefore, $(zw)^* = rse^{-i(\theta+\phi)} = (re^{-i\theta})(se^{-i\phi}) = z^*w^*$. If $z = re^{i\theta}$, then $1/z = 1/(re^{i\theta}) = (1/r)e^{-i\theta}$, so $(1/z)^* = (1/r)e^{i\theta}$. Finally, $1/z^* = 1/(re^{-i\theta}) = (1/r)e^{i\theta} = (1/z)^*$.

(e) If $z = \dfrac{a}{b+ic}$, then $|z|^2 = zz^* = \dfrac{a}{b+ic}\left(\dfrac{a}{b+ic}\right)^* = \dfrac{a}{b+ic}\cdot\dfrac{a}{b-ic} = \dfrac{a^2}{b^2+c^2}$.

5.36 ⋆⋆ From Eqs.(5.64), (5.65), and (5.70) we can calculate the various constants. A and δ are the same as in Example 5.3 (changing the initial condition doesn't affect them), but $B_1 = 0.945$ and $B_2 = 0.0429$. The resulting graph is the solid curve shown. The dashed curve is that of Fig.5.15(b). With different initial conditions, the two curves differ at first, but after a couple of cycles the transients have pretty well died out and the two graphs are indistinguishable.

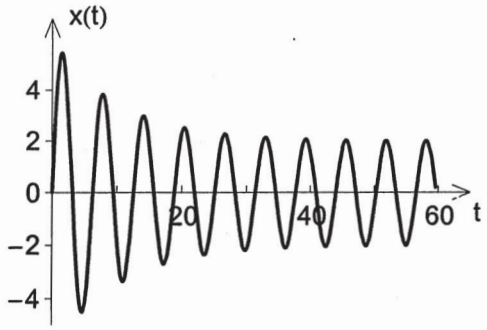


5.37 ⋆⋆ With the help of Eqs.(5.64), (5.65), and (5.70) we find that $A = 26.9$, $\delta = 3.04$, $B_1 = 26.7$, and $B_2 = -6.18$. The corresponding motion is shown by the solid curve, while the dashed curve is the transient alone.



The most obvious difference between the present case and that of Example 5.3 is that here $\omega_o < \omega$, so that the transient imposes a slow rise and fall on the driven oscillations. In Fig.5.15 of Example 5.3, where $\omega_o > \omega$, the transient superposes a rapid oscillation on the driven motion. A subtler difference is that in Fig.5.15 where $\omega_o > \omega$ the phase shift δ is close to zero and the motion is in step with the driving force. Here, with $\omega_o < \omega$, $\delta \approx \pi$ and the motion is nearly 180° out of phse with the driving force. [The driving force has maxima at integer values of t whereas $x(t)$ has minima.]

5.38 ⋆⋆ With the given parameters, the constants in (5.69) are easily found. Given that $\omega = \omega_o$, (5.65) tells us that $\delta = \pi/2$ and (5.64) that $A = f_o/2\beta\omega = 2$. Thus (5.70) implies that $B_1 = 0$ and $B_2 = (v_o - \omega A)/\omega_1$, or, since $\omega_1 = \sqrt{\omega_o^2 - \beta^2} = 0.995$, $B_2 = 4.02$. The graph of (5.69) with these values of the constants is as shown. Initially the oscillations have a large amplitude (somewhere about 5) but after a few cycles, the transients have died out and the motion is indistinguishable from the "attractor" with amplitude $A = 2$.



5.39 ⋆⋆ See Figure 5.15.

5.40 ⋆ We can save ourselves a little trouble if we note that A is maximum if and only is A^2 is maximum, which occurs if and only if $1/A^2$ is minimum. In fact, let's consider $(f_o/A)^2 = (\Omega - \omega_o^2)^2 + 4\beta^2\Omega$, where I'll use the variable $\Omega = \omega^2$. To find the minimum of this quantity we have only to differentiate and set the derivative equal to zero:

$$\frac{d}{d\Omega}\left(\frac{f_o^2}{A^2}\right) = 2(\Omega - \omega_o^2) + 4\beta^2,$$

which vanishes when $\Omega = \omega_o^2 - 2\beta^2$, that is, $\omega = \sqrt{\omega_o^2 - 2\beta^2}$. It is easy to check that the second derivative is positive. Therefore $(f_o/A)^2$ is minimum and A is maximum as expected.

5.41 ⋆ Provided β is significantly less that ω_o, the maximum of A^2 comes when $\omega \approx \omega_o$ and, at this point, the denominator of Eq.(5.71) is approximately $4\beta^2\omega_o^2$. Thus A^2 is equal to half its maximum when the denominator is equal to $8\beta^2\omega_o^2$, or when $(\omega^2 - \omega_o^2)^2 = 4\beta^2\omega_o^2$. This simplifies to $(\omega - \omega_o)(\omega + \omega_o) = \pm 2\beta\omega_o$. Since $(\omega + \omega_o) \approx 2\omega_o$, this says that the half maximum occurs at $\omega = \omega_o \pm \beta$.

5.42 ⋆ The period of the pendulum is $\tau = 2\pi\sqrt{l/g} = 10.99$ s. Therefore the quality factor is $Q = \pi(\text{decay time})/\tau = \pi \times (8\text{ h})/(10.99\text{ s}) \approx 8,000$.

5.43 ⋆⋆ **(a)** Assuming that the weight of the four men is evenly distributed among the four springs, we can substitute $m = 80$ kg and $x = 2$ cm into the equation $mg = kx$ for any one spring. This gives $k = mg/x = 80 \times 9.8/0.02 \approx 4 \times 10^4$ N/m.

(b) Each axle assembly is attached to two springs, so the effective spring constant of its support is $2k$ and its natural frequency is

$$f = \frac{1}{2\pi}\sqrt{\frac{2k}{m}} = \frac{1}{2\pi}\sqrt{\frac{2 \times (4 \times 10^4)}{50}} \approx 6\text{ Hz}.$$

(c) If the distance between bumps is d, then $v = fd = 6 \times 0.8 \approx 5$ m/s or roughly 10 mi/h.

5.44 ⋆⋆ **(a)** Since $x = A\cos(\omega t - \delta)$, the total energy is

$$E = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}kx^2 = \tfrac{1}{2}m\omega^2 A^2\cos^2(\omega t - \delta) + \tfrac{1}{2}kA^2\sin^2(\omega t - \delta).$$

Because $\omega \approx \omega_o$, we can replace $k = m\omega_o^2$ by $m\omega^2$, and then, since $\cos^2\theta + \sin^2\theta = 1$, we get $E = \frac{1}{2}m\omega^2 A^2$, as claimed.

(b) The rate at which the damping force dissipates energy is $F_{\text{dmp}}v = bv^2 = 2m\beta v^2$. Therefore the energy dissipated in one period is

$$\Delta E_{\text{dis}} = \int_0^\tau 2m\beta v^2 dt = 2m\beta\omega^2 A^2 \int_0^\tau \sin^2(\omega t - \delta)dt.$$